

Fraud × Prevention

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Introduction

Objective : How data-driven analysis can improve fraud detection and prevent losses caused by fraud cases at AAJ Bank

Problem Statement : AAJ Bank faces rising fraud cases, causing financial losses and losing trust, the challenge is to use advanced analytics, machine learning, and appropriate data to detect and prevent fraud effectively.

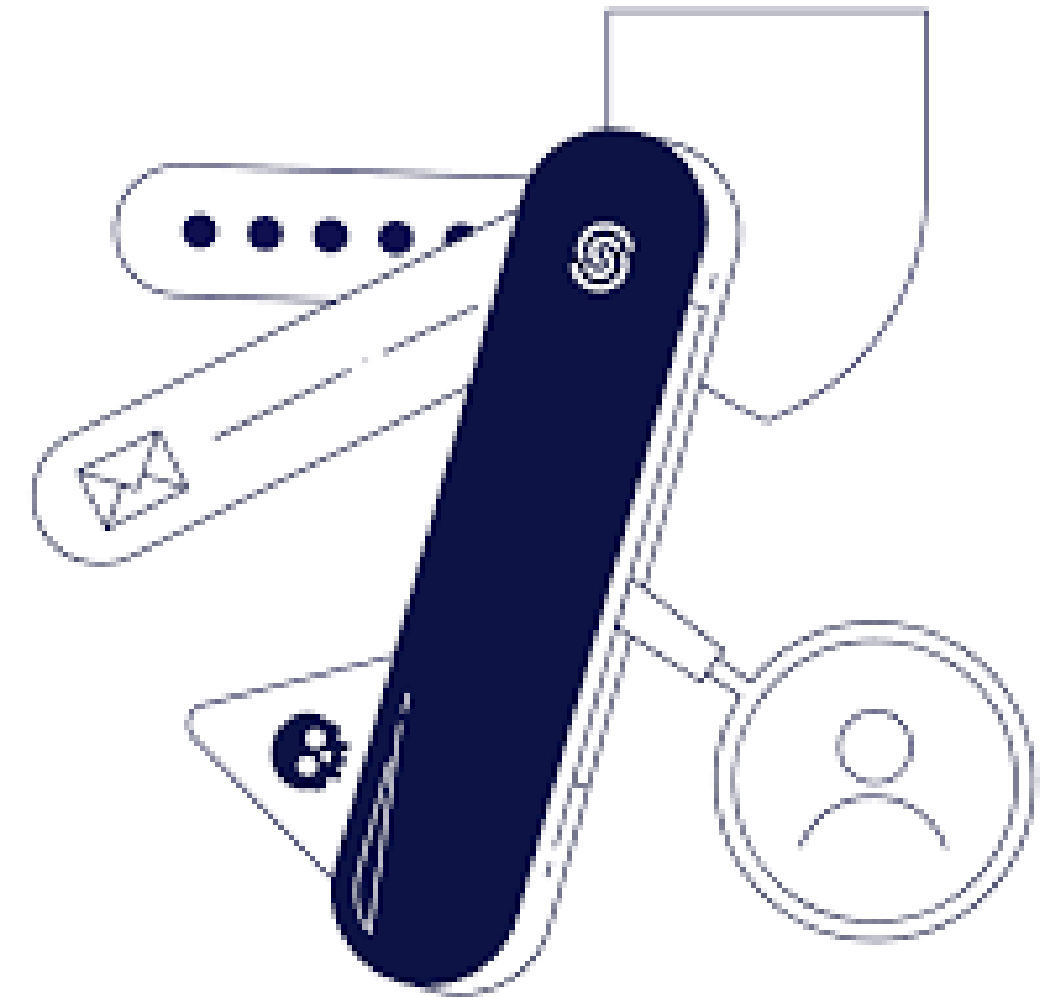


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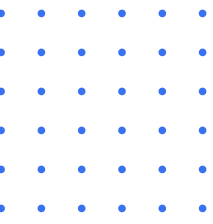
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Explanatory Variables

1. Transaction Anomalies
2. Account Activity Logs
3. Customer Profiles
4. Behavioral Patterns
5. Historical Fraud Cases



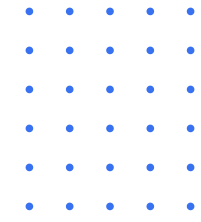
Transaction Anomalies



Transaction anomalies refer to unusual or unexpected patterns in customer transactions such as sudden large withdrawals, frequent transfers in larger amount or unusual location-based transactions

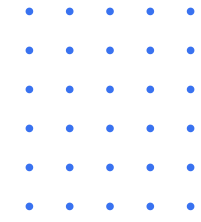
- Machine Learning (ML) : Algorithms like supervised learning and unsupervised clustering can learn from historical data to identify unusual or fraudulent patterns in real time
- Real-Time Transaction Monitoring Systems : Sending instant notifications when something unusual or different from the norm happens
- Cloud-Based Fraud Detection Tools (AWS Fraud Detector) : Reduce online payment fraud by stopping suspicious online payment transactions before processing

Account Activity Logs



Account activity logs capture detailed records of all transactions and account interactions. Analyzing these logs helps identify irregular activities such as multiple failed login attempts, changes to account details without proper verification

- SQL : Use Query and manage large datasets to extract relevant transaction information.
- Python : Utilize libraries (Pandas and Sklearn) for data manipulation, analysis, and machine learning
- Apache Spark : Using Spark Streaming and machine learning AAJ bank can spot unusual patterns as they happen. This allows for instant alerts or 2 factor authentication



Customer Profiles

Customer profiling means collecting and studying information about a customer's behavior, background, and past transactions. This helps the bank spot any strange or suspicious activities that could be signs of fraud

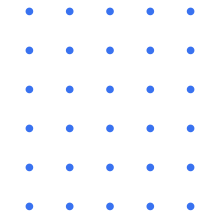
- Machine Learning (ML) : Helps detect changes in spending patterns by learning from historical customer behavior.
- Data Analytics Platforms (SQL, Python, Power BI): Process and visualize customer data, allowing for deeper insights.
- Customer Relationship Management (Salesforce) : Store detailed customer information, including transaction history and interactions.

Behavioural Pattern

Behavioral patterns involve analyzing real-time customer transaction activities, such as sudden spikes in transactions or odd transaction times. By monitoring these behaviors, the bank can quickly identify suspicious actions that deviate from a customer's usual behavior

- Real-Time Analytics (e.g., Apache Kafka, Apache Spark): Processes transaction data instantly, helping to detect anomalies like rapid multiple transactions.
- Artificial Intelligence (AI): AI algorithms analyze behavioral patterns to identify unusual transaction activity in real-time.
- Device Fingerprinting Tools (e.g., ThreatMetrix): Tracks and verifies devices used for transactions, identifying suspicious device usage.

Historical Fraud Cases



Analyzing historical fraud cases helps banks improve their fraud detection systems by recognizing patterns in previous fraud attempts. This analysis provides insights into the types of fraud that may occur, common indicators, and areas where systems need improvement

- Data Mining (RapidMiner, SAS): Extracts patterns from large datasets of past fraud incidents, identifying trends and common fraud indicators.
- Fraud Detection Systems (AI/ML): Continuously improve by learning from previous fraud cases, reducing the likelihood of false positives and negatives.
- Predictive Analytics (Python) : Forecasts future fraud trends based on patterns detected in historical fraud data.



Solutions

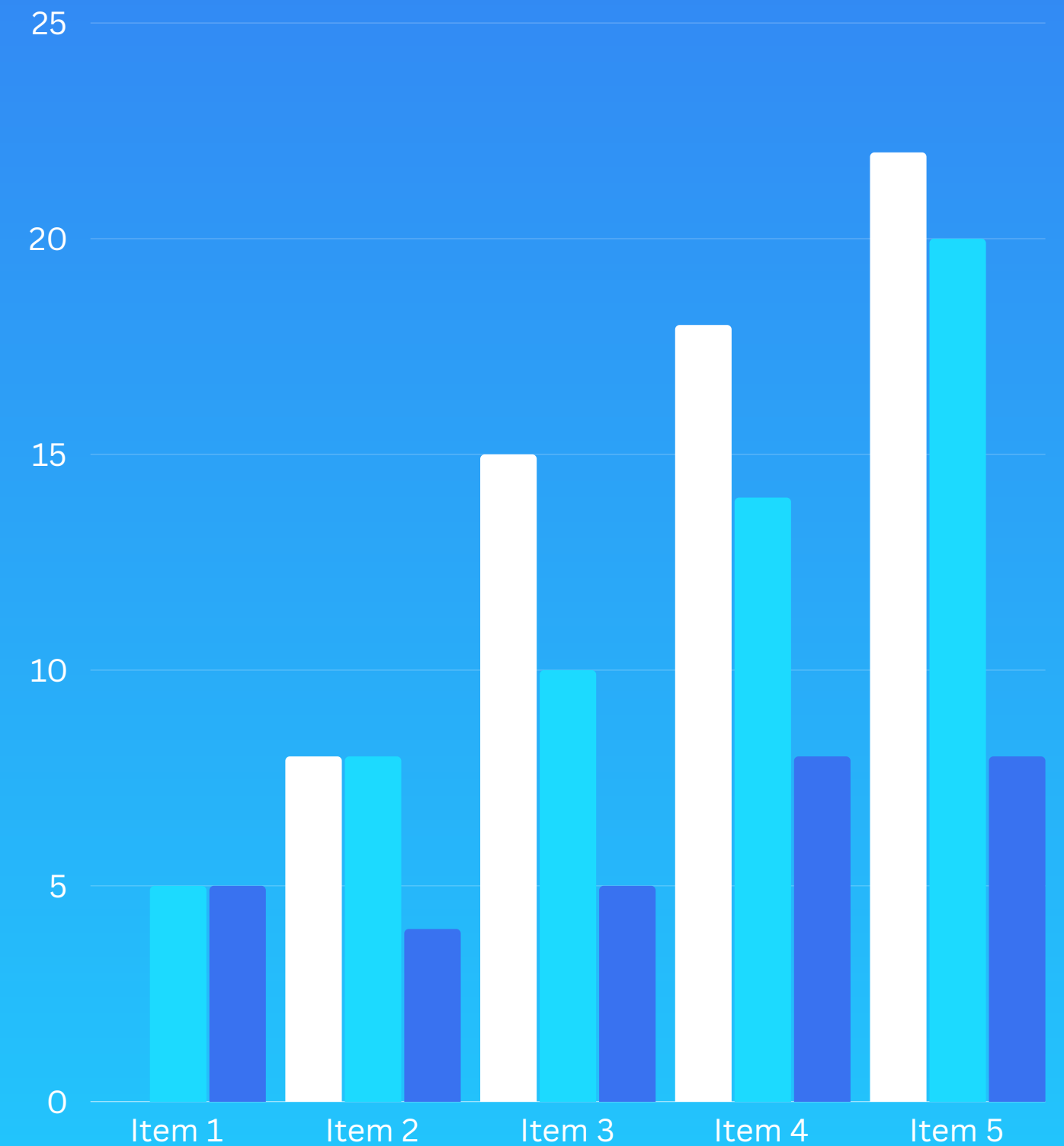
Non-ML methods

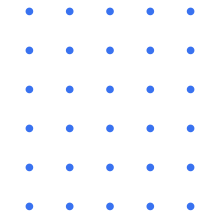
1. Rule based system
2. Statistical Analysis - Descriptive Statistics, Threshold Analysis
3. Geographical and Temporal Analysis
4. Heuristic based approaches

ML methods

Supervised ML Algorithms

1. Logistic Regression
2. Random Forest

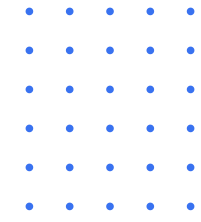




Advantages of Non-ML Methods

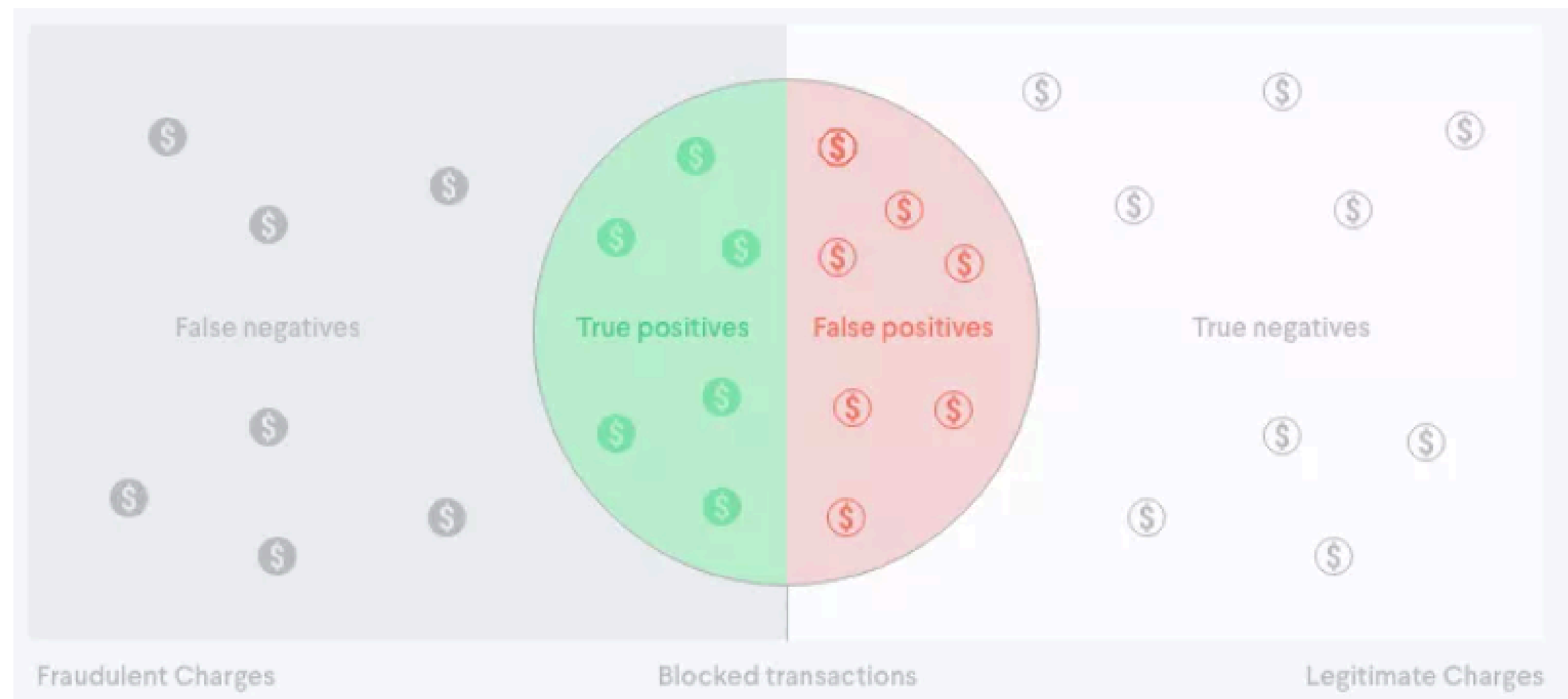
1. Simplicity and Ease of Implementation
2. Interpretability
3. Low computational Efforts
4. Immediate Response





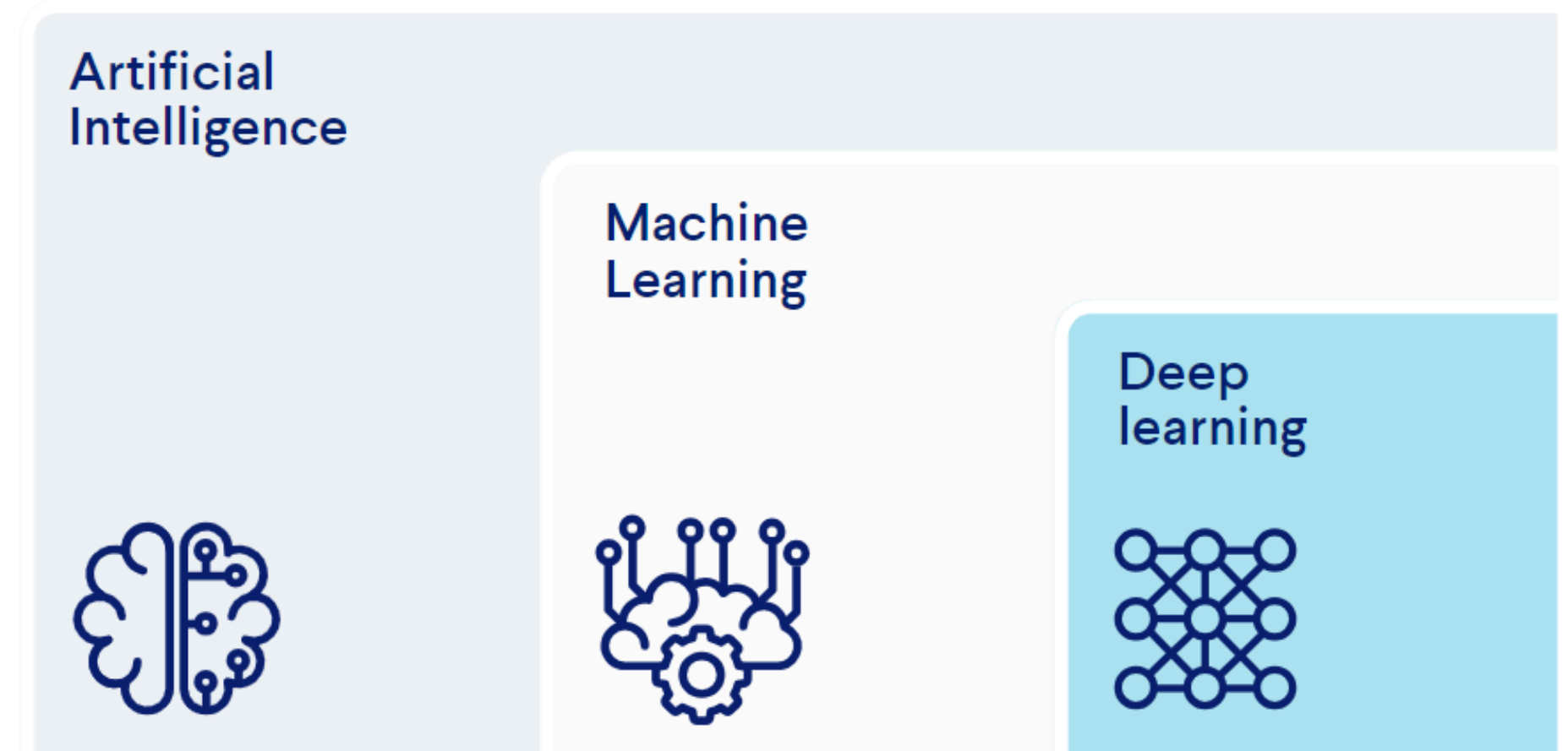
Drawbacks of Non-ML Methods

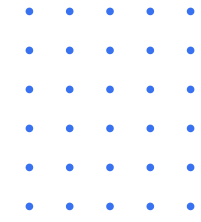
1. Limited Adaptability
2. High False positives
3. Maintenance Challenges
4. Inability to handle complex problems



Why Machine Learning ?

1. Anomaly detection
2. Risk Scoring
3. Network Analysis
4. Text Analysis
5. Identity Verification
6. Adaptive Learning
7. Faster and more efficient detection
8. Reduced manual review time
9. Better prediction with large datasets
10. Cost effective solution





Explanatory Variables

Device Fingerprinting

1. Device model
2. Operating system
3. IP Address
4. Device location
5. Frequency of device use
6. Number of accounts linked to the device

Behavioural Biometrics

1. Typing speed
2. Swipe gestures
3. App usage patterns
4. Time of interaction

Account takeover Prevention

1. Number of failed login attempts
2. Login location
3. Device type
4. Time of login
5. IP address
6. New or unusual login patterns



Friendly Fraud Detection

1. Customer purchase history
2. Return rates
3. Chargeback patterns
4. Time of claims
5. Product delivery confirmation

Invoice Fraud Detection

1. Invoice amount
2. Vendor details
3. Duplicate invoices
4. Discrepancy in amounts
5. invoice submission frequency

Loyalty Program Fraud Detection

1. Points accumulated rate
2. Redemption patterns
3. Account activity frequency
4. Unusual redemption locations
5. Account balance charges

Machine Learning Pipeline



Exploratory Data Analysis



Data Cleaning and Transformation



Data Visualization



Model Training



Model Evaluation



Dataset

<https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>

Name of the variable	Description
step	Maps a unit of time in the real world. 1 step is 1 hour of time.
type	Indicates the type of transaction. This can be CASH-IN, CASH-OUT, DEBIT, PAYMENT or TRANSFER
amount	amount of the transaction in local currency
nameOrig	identifier of the customer who started the transaction
oldbalanceOrg	initial balance of the originator before the transaction
newbalanceOrg	originator's balance after the transaction
nameDest	identifier of the recipient who received the transaction
oldbalanceDest	initial balance of the recipient before the transaction
newbalanceDest	recipient's balance after the transaction
isFraud	indicates whether the transaction is actually fraudulent or not. The value 1 indicates fraud and 0 indicates non-fraud

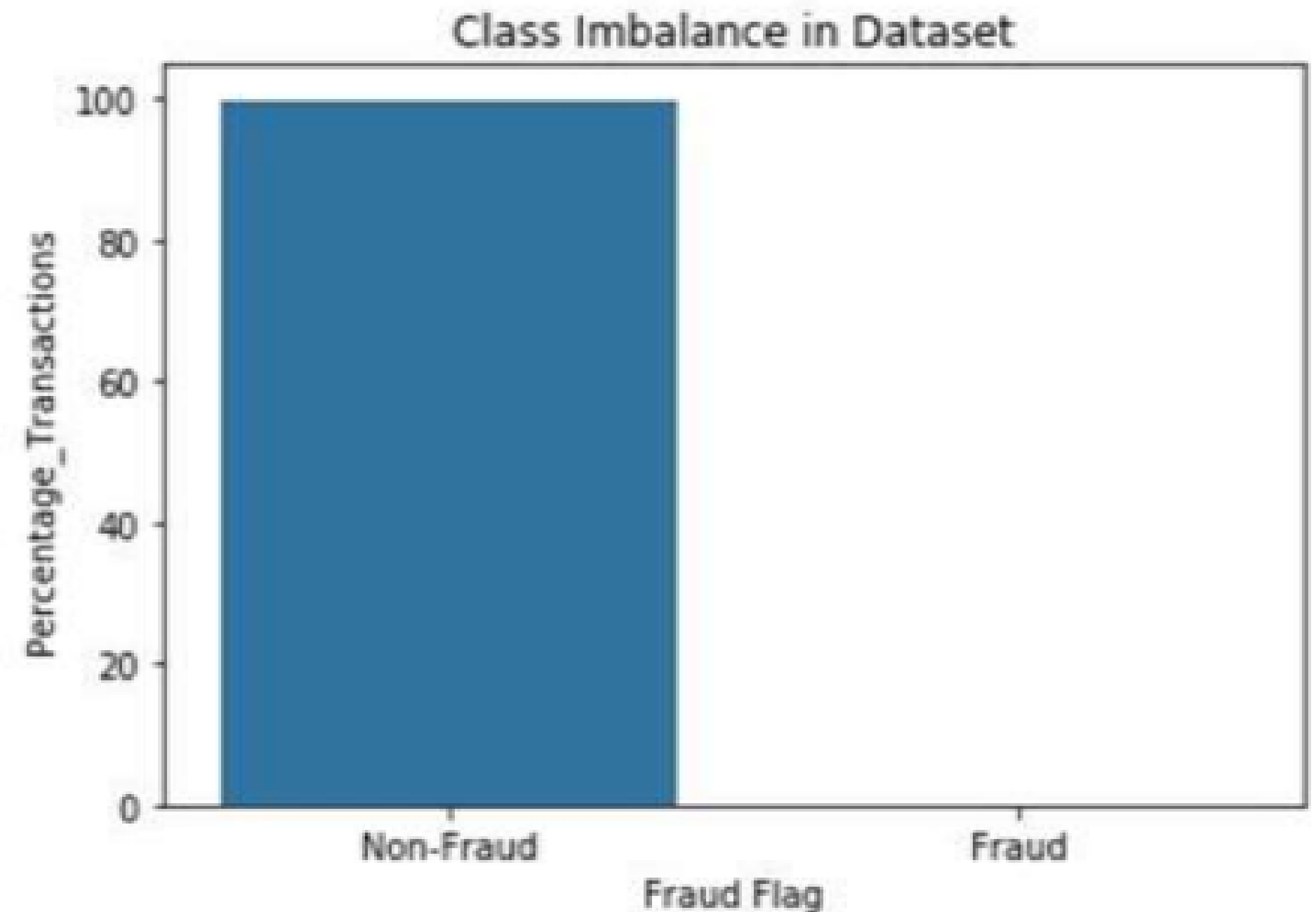


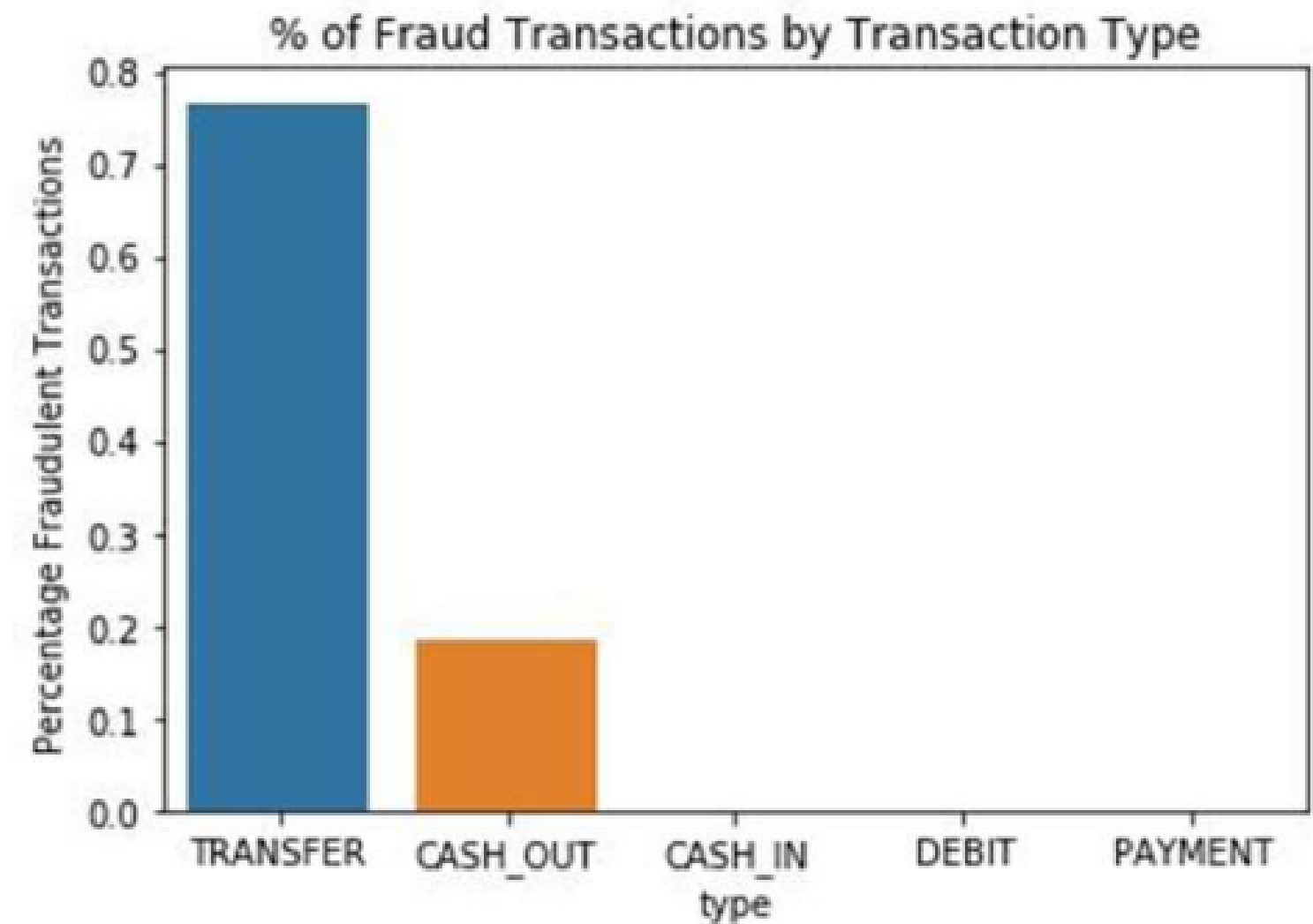
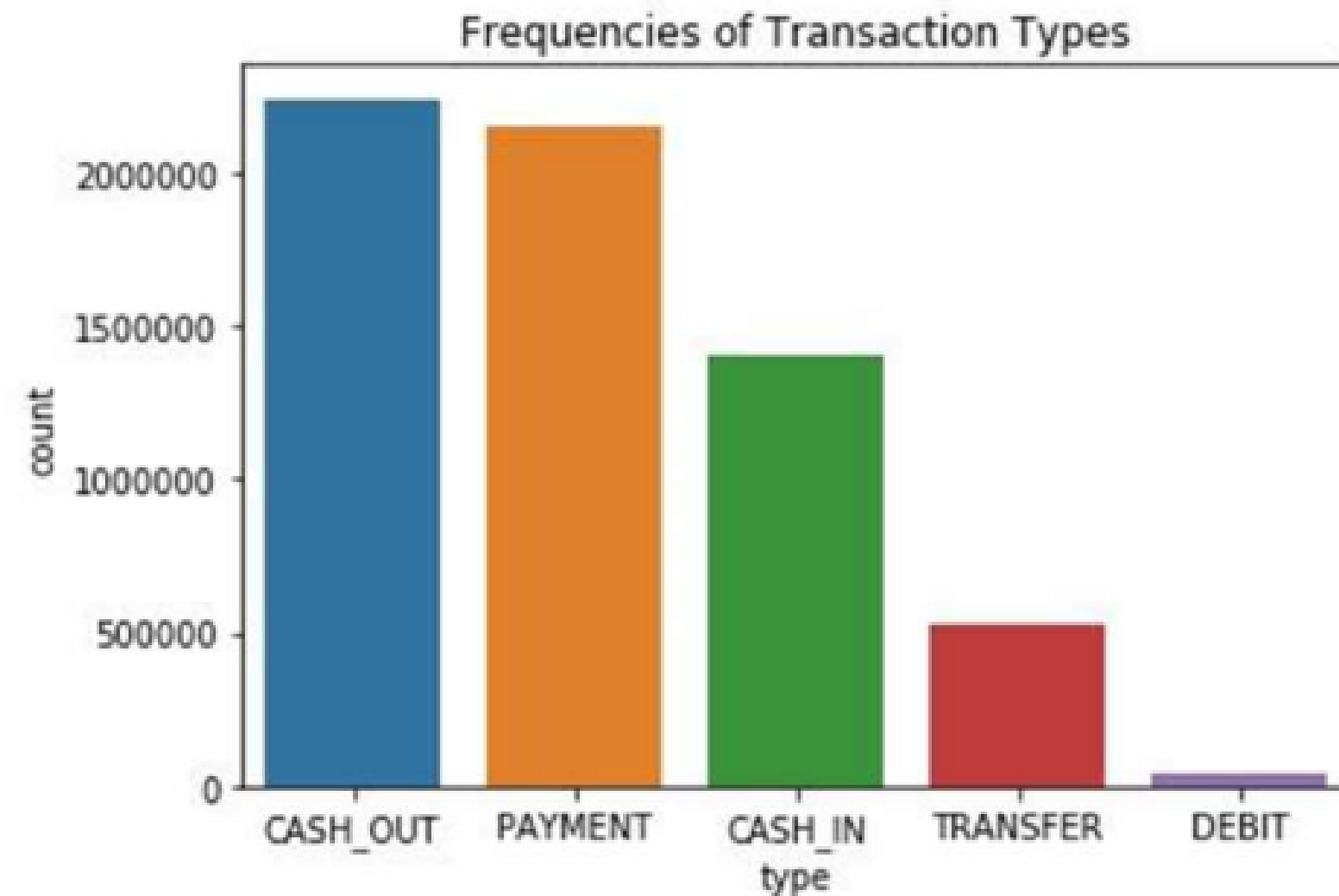
Imbalanced Data

We can tackle the issue of imbalanced dataset with the following approaches:

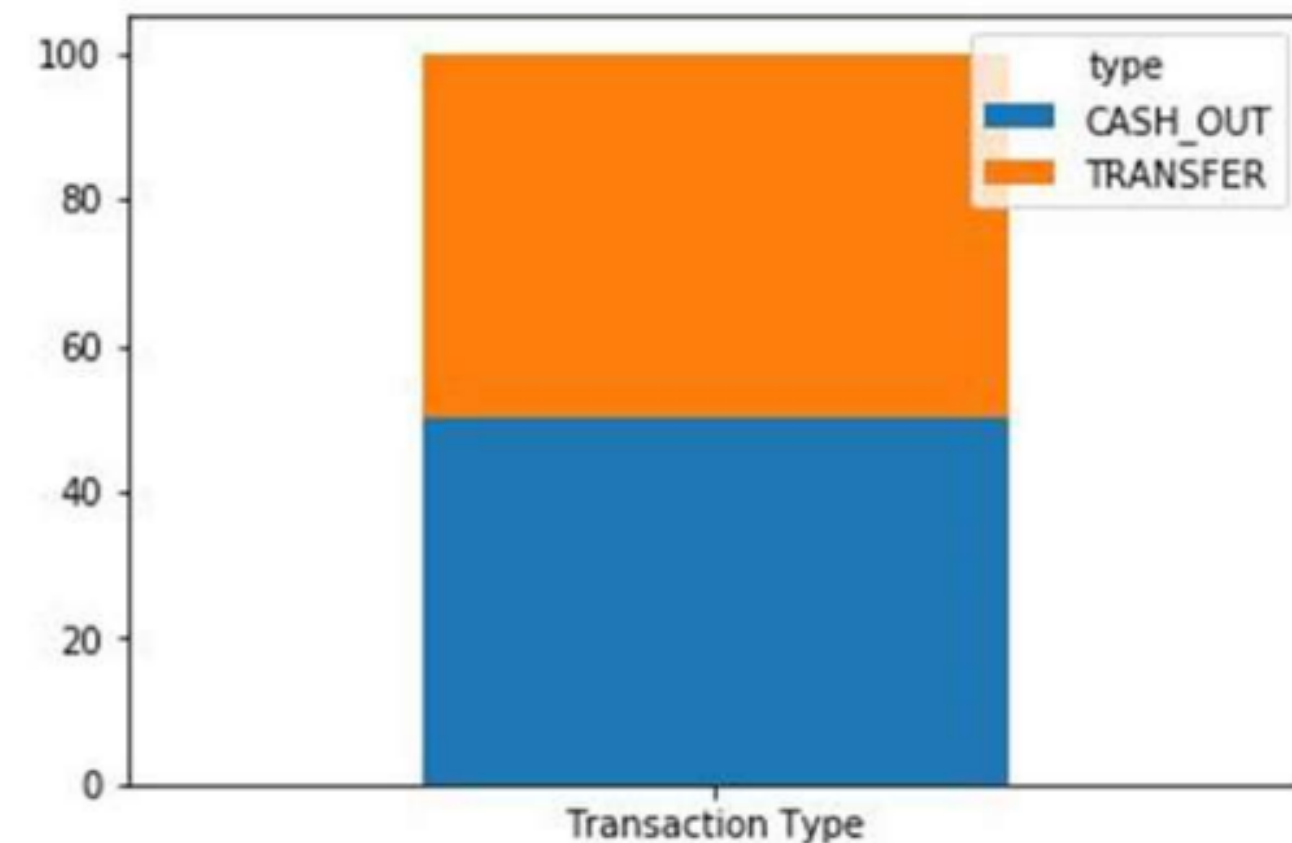
1. SMOTE
2. Class Weight based Approaches
3. Alternative Methods

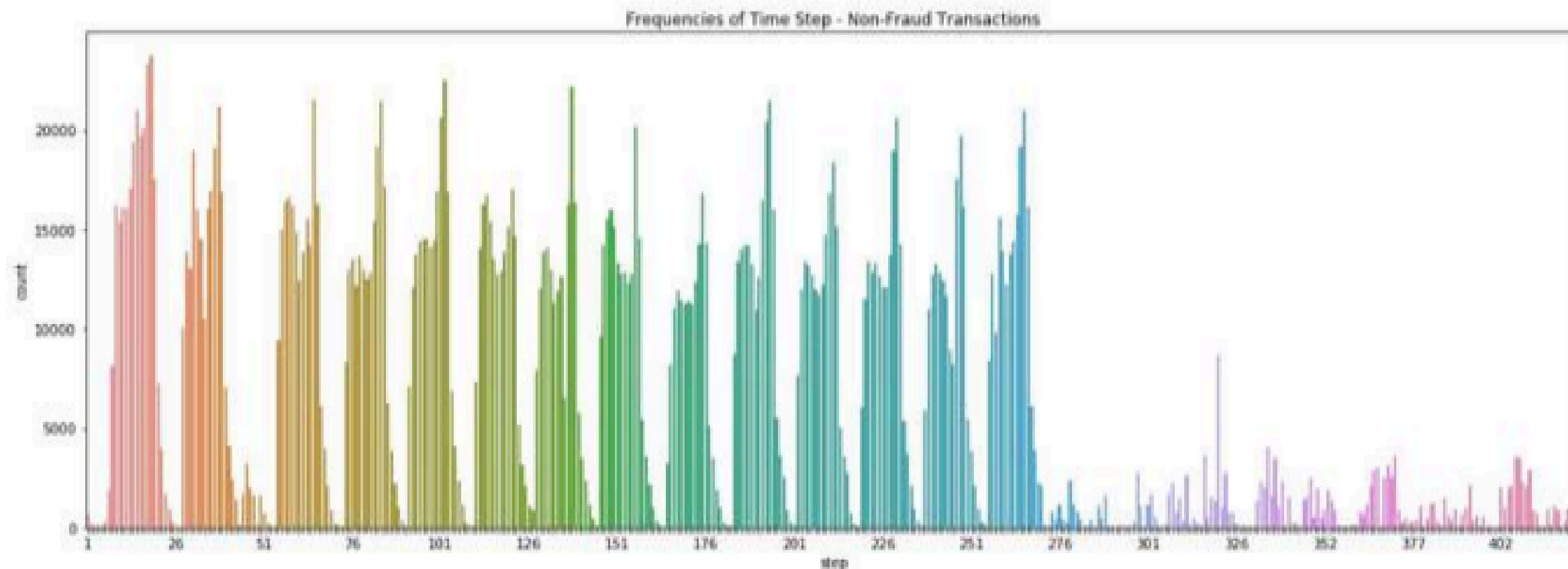
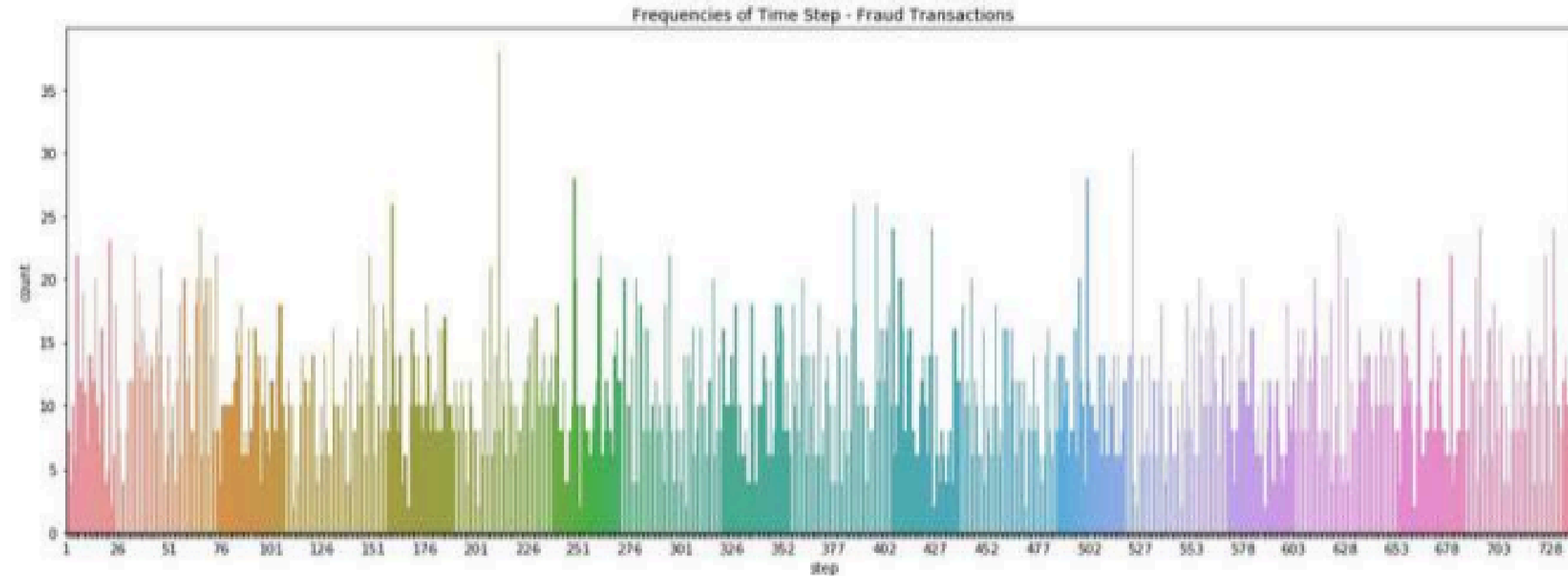
	Fraud Flag	Percentage_Transactions
0	Non-Fraud	99.87
1	Fraud	0.13



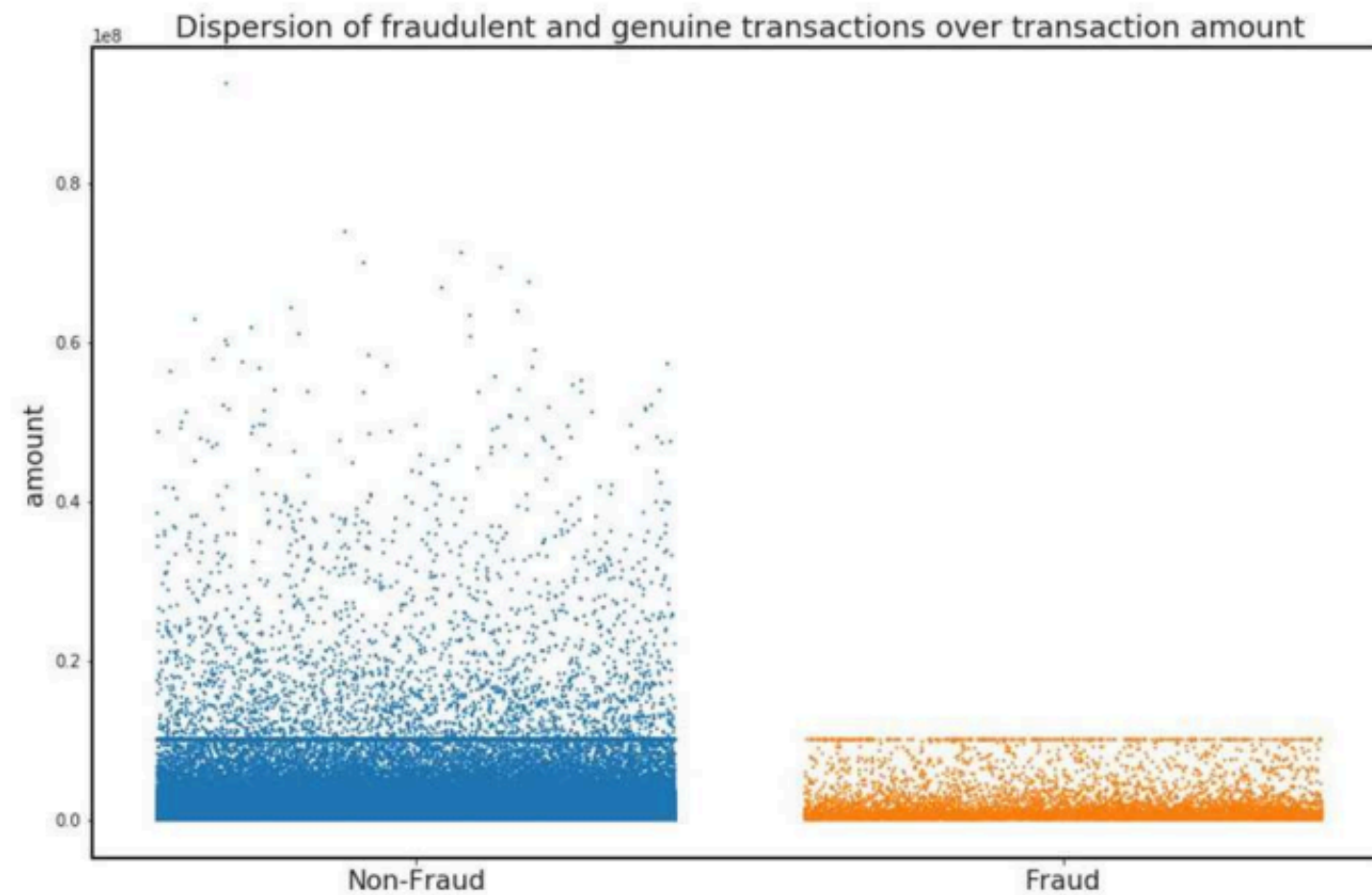


1. The most frequent transaction types are CASH-OUT and PAYMENT.
2. Only CASH-OUT and TRANSFER transactions can be fraudulent. So, we will retain only these two types of transactions in our dataset.
3. There is almost equal likelihood that a fraudulent transaction can be CASH-OUT and TRANSFER.

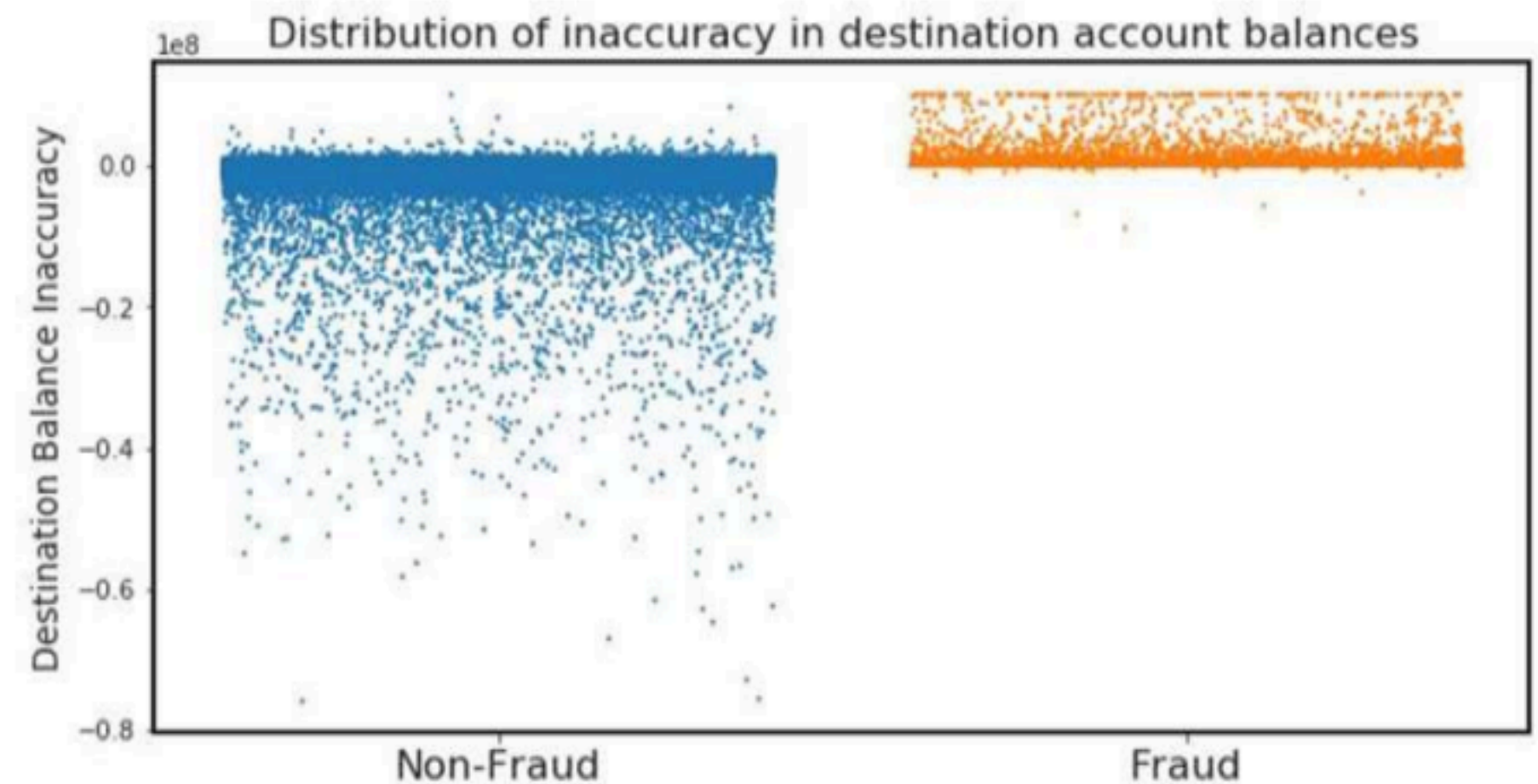
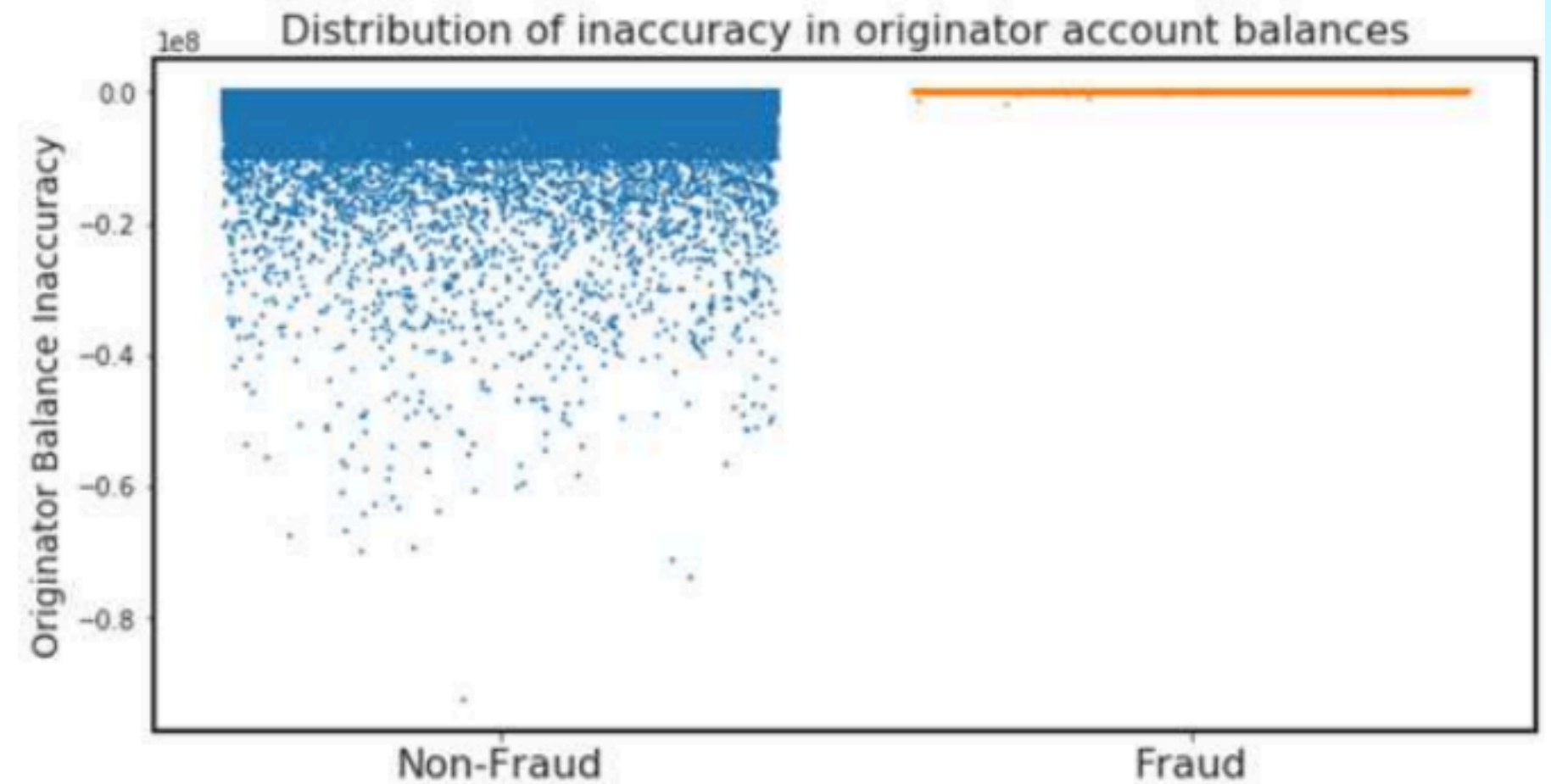




Fraudulent transactions are almost uniformly spread out across time steps, whereas non-fraud transactions are more concentrated in specific time steps.

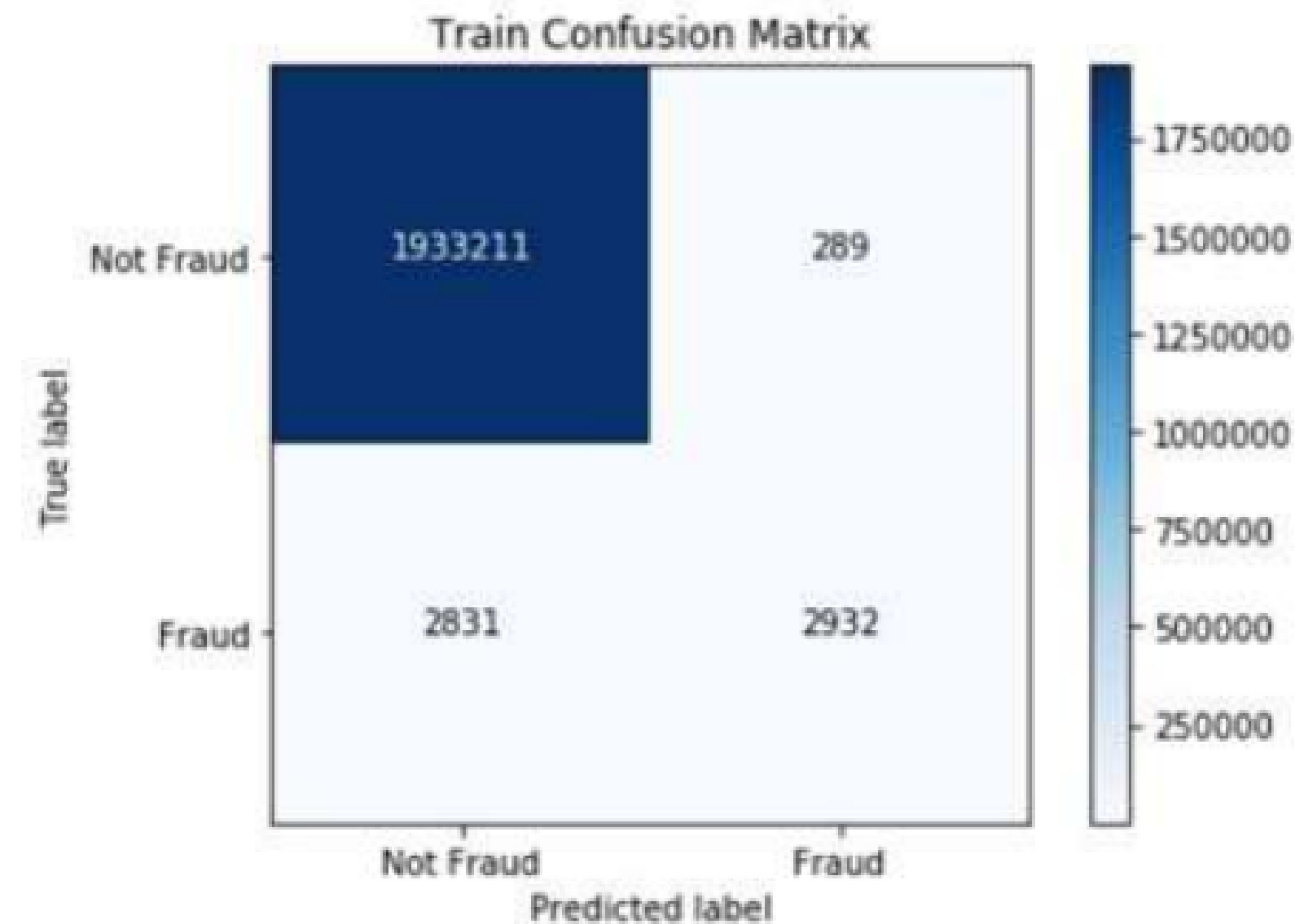


It appears that the inaccuracy in destination balance is almost always negative for non-fraud transactions, whereas it is almost always positive for fraud transactions.

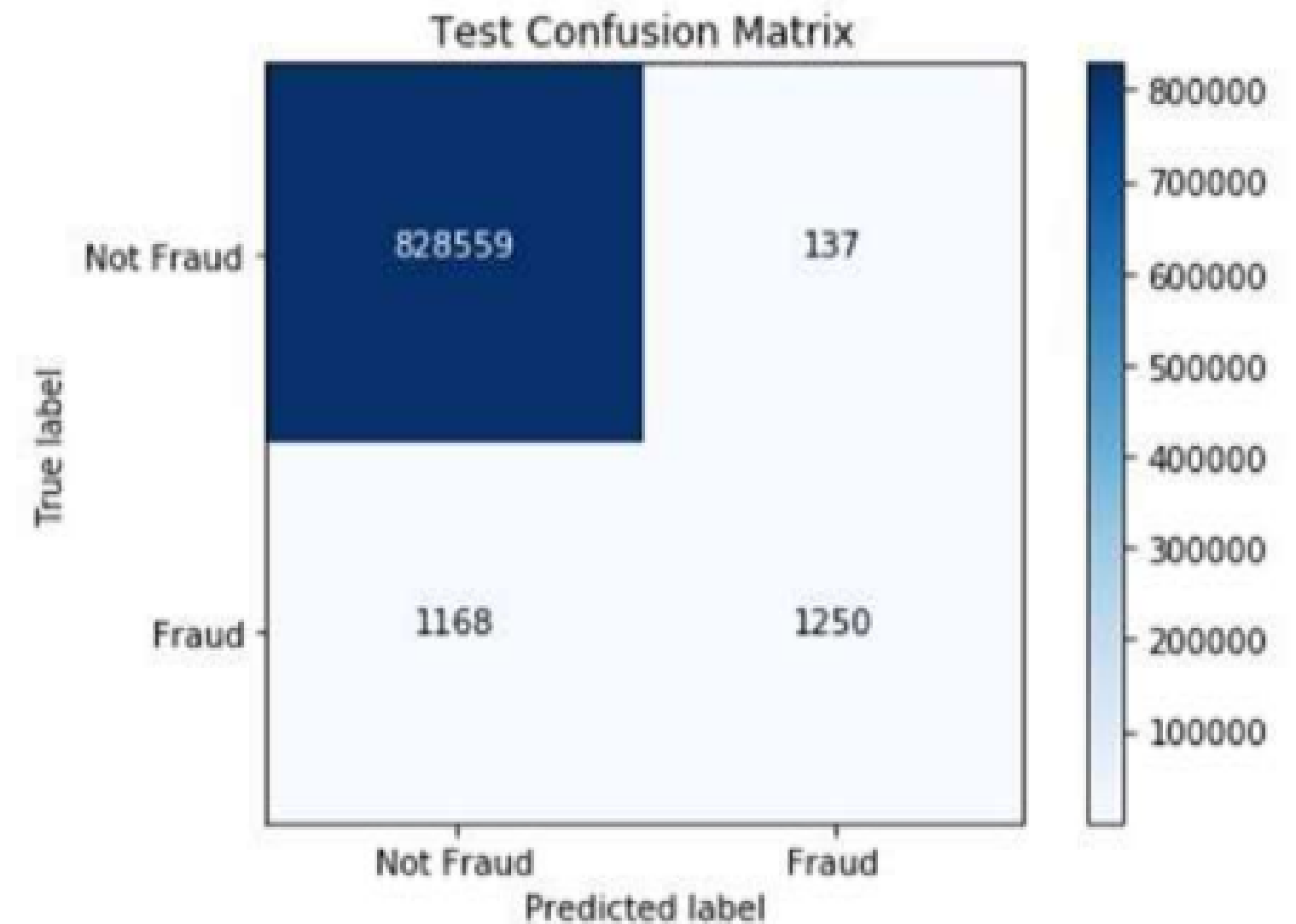


Logistic Regression

Precision: 91.03%
Recall: 50.88%

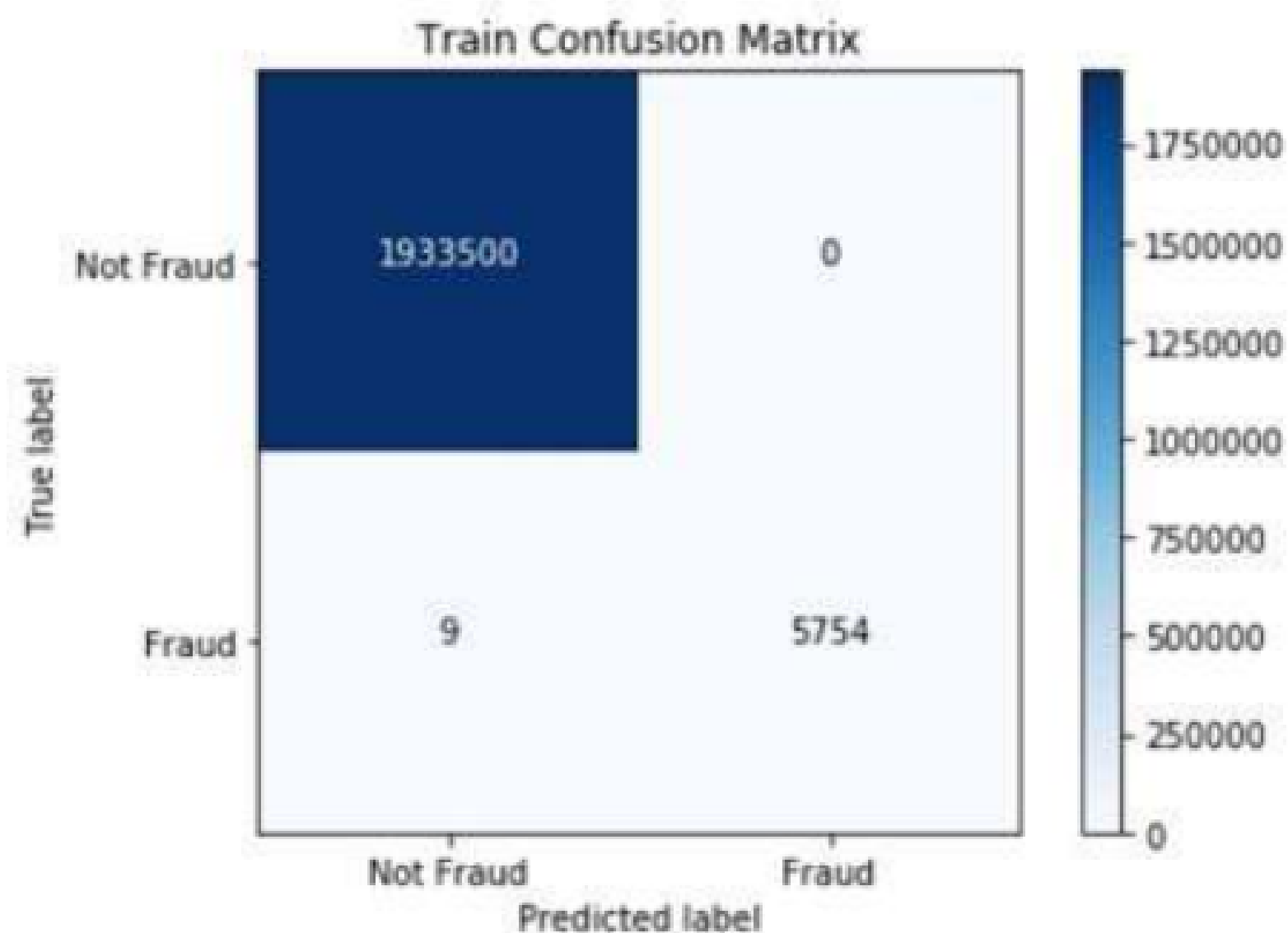


Precision: 90.12%
Recall: 51.7%

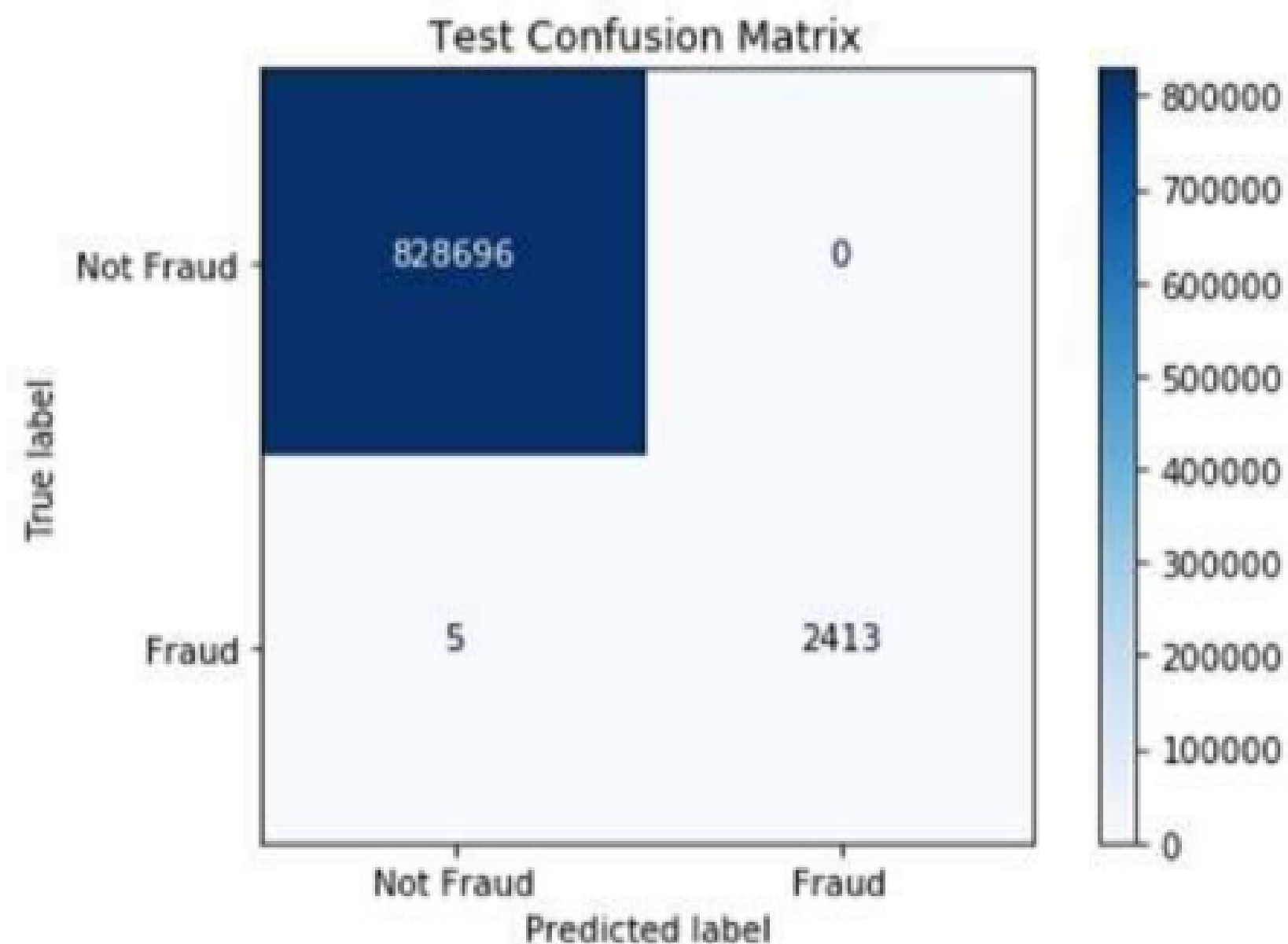


Random Forest

Precision: 100.0%
Recall: 99.84%



Precision: 100.0%
Recall: 99.79%



Proposed Solution +

At the point of the transaction, the model gives each customer a risk score on the scale of 1-100. The higher the score, the higher the probability of fraud.

You can choose what level of risk is right for your business, and set thresholds for what proportion of transactions you want to allow, block, and manually review.



Impact on Operations

- Enhanced Reputation : A robust but user-friendly fraud detection mechanism boosts the company's market reputation
- Fewer Chargebacks : Decreases in unauthorized transactions lead to fewer customer disputes
- Cost Savings - Automation reduces the number of employees required for fraud checks
- Faster Processing : Transactions are flagged automatically, reducing the need for human interference



References

- <https://cs229.stanford.edu/proj2018/report/261.pdf>
- <https://stripe.com/en-ca/resources/more/how-machine-learning-works-for-payment-fraud-detection-and-prevention#supervised-learning>
- https://www.researchgate.net/publication/374083997_FRAUD_DETECTION_USING_MACHINE_LEARNING
- <https://seon.io/resources/fraud-detection-with-machine-learning/>
- <https://www.ravelin.com/insights/machine-learning-for-fraud-detection>
- https://github.com/jahnavish/Fraudulent_Transactions_Detection/blob/main/fraud_transactions.py





Thank You

